

Employing instantaneous positive sequence symmetrical components for voltage sag source relative location



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ABSTRACT

Modern industrial facilities are vulnerable to voltage sags and short interruptions as they were in the past to sustainable interruptions. Therefore, modern reliability assessment should include voltage sags and short interruptions. Hence, in order to improve power system reliability it is necessary to mitigate these voltage variations. One of the first steps towards this objective is to locate the source of these disturbances. This paper investigates a number of methods for sag source relative location. Said methods are based on different electrical parameters. Using instantaneous positive sequence components, five novel extended methods for voltage sag source location are proposed for accommodating the transient nature of voltage sag events and for improving unsuitable operation of methods on asymmetrical sags and earth faults. Due to the fact that short circuits are the main reason resulting in voltage sags, this paper focuses on the location of voltage sag sources due to line faults. A comparative analysis between the existing methods and the five novel extended methods is performed by applying extensive numerical simulations in a Brazilian regional transmission and sub-transmission utility. Finally, the effectiveness of all methods is obtained. The results show that the novel extended methods are more effective and among them, the extended methods based on reactive power can locate sag sources with an accuracy of 88%. Also, implementation of the novel extended methods on signal processors is easier with respect to the already implemented methods.

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1. Introduction

Voltage sags are an important power quality disturbance that can occur in utilities or in the end user with a frequency of even several thousand times per year [1]. This phenomenon can also cause production loss in a typical industrial plant [2–4]. Therefore, they have a great influence in industrial plant reliability. The magnitude and duration of these voltage sags can be between 10% and 90% of nominal voltage and half cycle to one minute, respectively. Voltage sags can be symmetrical or asymmetrical and may be caused by starting of large motors, transformer energizing, capacitor switching, as well as by overloads and faults. Voltage sags caused by faults are not confined to the fault point, and are propagated through the system affecting connected loads far away from the fault location, causing many disagreements between utilities (transmission–sub transmission, sub transmission–distribution,

distribution–distribution and distribution–micro grids) and also between utilities and consumers. Therefore, identification of the event causing the sag is crucial so as to solve such issues. Voltage sag source location is an important topic and is the first stage toward mitigation of power quality problems [5,6].

Many studies and methods have already been reported [7–35] on the subject of voltage sag source location. These methods were divided into five categories. The first category is based on “disturbance power and energy” [7,8–13]. Leborgne et al. [14,15] investigated an alternative approach based on power flow. The second category is based on changes of current. Li et al. [16] introduced a method based on the polarity of slope of system trajectory at the monitoring points. The polarity of active current was proposed to classify the sag source [17]. The third category focuses on impedance changes. Method [16] was generalized by the same authors of the “resistance sign” approach [18]. In this method, the sign of the resistance obtained from incremental impedance was the criterion utilized to locate the sag source. The results of the method depended on the data cycles, so another method [19] was introduced based on negative sequence components. Later, Zhu et al. [20] determined the sag source applying the same theory, according to the neutral point operation mode. Yi et al. [21] pre-

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Nomenclature

PQM	Power quality monitor
DS	Downstream
US	Upstream
DPE	Disturbance power and energy
RP	Reactive power
RCC	Real current component
SST	Slope of system trajectory
DR	Distance relay
(.)	Number of equation relation to the rule of methods
(.) _{sag} , (.) _{presag}	During sag and pre-sag
$\Delta(.) = (.)_{sag} - (.)_{presag}$	Change due to the sag
$\angle(.)$	Phase angle
V, I	Voltage and current phasor
θ	Phase angle between voltage and current
Z, R	Impedance and resistance
(.) ⁺ , (.) ⁻ , (.) ⁰	Positive, negative and zero sequence components
v, i, z, r	Instantaneous voltage, current, impedance, resistance
$\Delta P, \Delta Q$	Active and reactive disturbance power
ΔE	Disturbance energy on ΔP
$p^+ = v^+ i^+ \cos(\theta)^+$	Active power obtained from instantaneous positive sequence components
$q^+ = v^+ i^+ \sin(\theta)^+$	Reactive power obtained from instantaneous positive sequence components
Δp^+	Disturbance active power obtained from instantaneous positive sequence components
Δq^+	Disturbance reactive power obtained from instantaneous positive sequence components
Δe^+_p	Disturbance active energy on Δp^+
Δe^+_q	Disturbance reactive energy on Δq^+
$(\theta)^+$	Phase angle between instantaneous positive sequence voltage and current

sented a method using the polarity of the real part of internal impedance in a single-port network. In an alternative method, the sag source was found using the impedance seen by distance relay [22]. This method was tested in a Brazilian network [23,24]. Later, the variation of angle was used instead of the impedance angle [25]. The fourth category is based on voltage measurements only. A method based on the voltage magnitude and phase-angle jump was proposed [26]. Another method [27] based on voltage magnitude at both sides of a power transformer was introduced. The fifth category locates the sag source according to current measurements. The first method in this category [28] is based on the phase change in the positive-sequence current phasor. Its application is, however, limited to systems supplied from two sides. Moradi et al. proposed a further method [29], which used magnitude change in the positive-sequence current phasor as an extra criterion to the method proposed by Pradhan et al. [28].

The testing of some discussed methods for voltage sag source location known from literature shows that in the cases of asymmetrical sags, these methods are rather ineffective [30]. Therefore, Polagzer et al. [8] generalized the methods based on positive sequence phasors. Moreover, since voltage sags are a transient power quality event, the methods based on phasor or positive sequence phasor might have questionable results. Thus, the same authors generalized the methods of Refs. [7,16–18] based on Clarke's components [9,31] in systems with DGs and presence of active loads, and improved the method of Ref. [28] as Refs. [32,33] (this will be discussed from a different viewpoint in Section of 4.2

(results obtained) in this paper). Also, Shao et al. [34] developed method [22] based on Park's components. Voltage sag source location methods along with their features were analyzed in the Catalan network [35]. Mohammadi et al. [36] used a robust machine learning approach, with very high effectiveness, in which they included features of all previous methods.

It is worth pointing out that magnitude, duration, recovery, prediction and detection are highly relevant alternative characteristics of voltage sags and should be taken into account besides the source location of voltage sags [37–39].

In this paper, the rules of six earlier methods for voltage sag source location are briefly explained. Also, five other methods that extended the former are introduced by using instantaneous symmetrical components. All the proposed extended methods are based on instantaneous voltages and currents and not on phasors. After showing the results of existing and extended methods, a comparative analysis is performed between all the methods by applying extensive numerical simulations in a Brazilian regional power network. In this network, six monitors and fourteen fault locations for which all type of faults (symmetrical, asymmetrical and earth faults with low and high faults resistances) at both upstream and downstream locations are simulated. The monitors in the network represent different kinds of system topology (i.e. radial, interconnected, one source, two sources and so on). The obtained effectiveness and accuracy of the discussed methods for all simulated cases is fully presented and a main comparison is performed between them.

2. Methods for voltage-sag source relative location

In methods for voltage-sag source relative location, the relative location of a sag source as observed from a power quality monitor (PQM) is defined as downstream (DS) or upstream (US), using the steady state active power flow direction as reference. DS is the region in the direction of the power flow and US is the region against it, as shown in Fig. 1. The methods for sag source relative location can be divided into several categories based on various criteria such as power, energy, current, voltage and impedance. A description of the conventional methods for DS/US location of voltage sag sources, which are used in this paper for generalization purposes, is given in Table 1.

3. Introducing extended methods for voltage-sag source relative location

In this section, the extended rules for existing original methods are obtained using instantaneous symmetrical components [32,40], and as the main contribution of this paper, novel methods are introduced.

3.1. Instantaneous symmetrical components transformation

Instantaneous voltages ($v_a(t)$, $v_b(t)$, $v_c(t)$) and instantaneous currents ($i_a(t)$, $i_b(t)$, $i_c(t)$) may be transformed into instantaneous positive, negative and zero-sequence voltages (v^+ , v^- , v^0) and currents (i^+ , i^- , i^0) in a-phase, where F denotes the Fortescue's

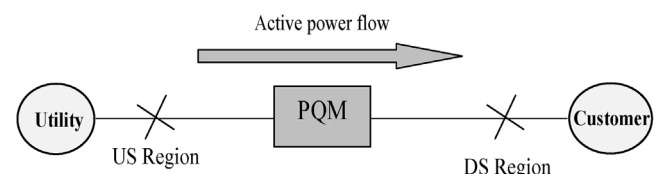


Fig. 1. Relative location of voltage sag source as US and DS.

Table 1
Description of the existing conventional methods for voltage-sag source location.

Criteria	Based on	Method	[Refs.]	Basic rule for voltage sags
Methods based on changes of power and energy: voltage sag sources can act like energy sinks. So, it can be assumed that the energy flow at the PQMs increases during DS voltage sags and decreases during US voltage sags. The criteria are mainly based on changes in instantaneous active and reactive powers and energies.	Instantaneous or active power	DPE (1)	[7]	$\begin{cases} \text{If } \Delta E(t) = \int_0^t \Delta P(u) du > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases}$
	Active and reactive power	RP (2)	[14]	$\begin{cases} \text{if } \Delta Q > 0 \text{ and } \tan(\theta) = \frac{Q_{\text{sag}}}{P_{\text{sag}}} > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases}$
Methods based on changes of current: voltage sags are a result of short-term increases in currents. Thus, currents measured at the PQMs increase during DS voltage sags and decrease during US voltage sags. The criteria are based on the slope of the system trajectory and the change in real current.	Line phasors	RCC (3)	[17]	$\begin{cases} \text{if first peak } \Delta(I \cos(\theta)) > 0 \Rightarrow \text{DS : for a,b,c phases} \\ \text{else} \Rightarrow \text{US} \end{cases}$
	Line phasors	SST (4)	[16]	$\begin{cases} \text{if slope } (I, V \cos(\theta)) \text{ line} < 0 \Rightarrow \text{DS : for a,b,c phases} \\ \text{else} \Rightarrow \text{US} \end{cases}$
Methods based on changes of impedance: the magnitude and angle of impedance change at the PQMs during voltage sags. The criteria are based on changes in the impedance estimated of a distance relay or resistance sign obtained of incremental impedance	Positive sequence phasors	DR (5)	[22]	$\begin{cases} \text{if } \Delta Z < 0 \text{ and } \Delta\angle Z_{\text{sag}} > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases}$
	Positive sequence phasors	DR (6)	[25]	$\begin{cases} \text{if } \Delta Z < 0 \text{ and } \Delta\angle Z > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases}$

transformation matrix, $(\cdot)^*$ denotes the complex conjugate and the “a” operator shifts the instantaneous vectors by an angle of 120° counter-clockwise, and the “a²” operator performs a 240° counter-clockwise phase shift [40]. The instantaneous positive and negative sequence components are complex conjugates, which are, $i^+ = i^{-*}$ and $v^+ = v^{-*}$, whereas the instantaneous zero sequence components i^0 and v^0 are always real.

$$\begin{pmatrix} i^+ \\ i^- \\ i^0 \end{pmatrix} = F \begin{pmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{pmatrix}, \begin{pmatrix} v^+ \\ v^- \\ v^0 \end{pmatrix} = F \begin{pmatrix} v_a(t) \\ v_b(t) \\ v_c(t) \end{pmatrix} \quad (7)$$

$$F = \frac{1}{3} \begin{pmatrix} 1 & a & a^2 \\ 1 & a^2 & a \\ 1 & 1 & 1 \end{pmatrix}, F^{-1} = (F^*)^T, a = 1 \angle 120^\circ = -\frac{1}{2} + j\frac{\sqrt{3}}{2} = e^{j2\pi/3}$$

An example of a DS voltage sag event due to an earth fault in a-phase can be seen in Fig. 2. The duration of the sag is 167 ms–280 ms. Magnitude and phase of instantaneous positive sequence voltage and current (Fig. 2c–f) are obtained from line instantaneous voltages and currents (Fig. 2a and b), which can be seen in the following form: $\Delta|v^+| < 0$, $\Delta|i^+| > 0$, $\Delta\angle v^+ = 0$ and $\Delta\angle i^+ < 0$.

3.2. Extension of DPE (1) method based on instantaneous positive sequence

By using p^+ in Eq. (1), the 1st novel extended method is introduced as the following rule (DPE (8)):

$$\begin{cases} \text{if } \Delta e_p^+ = \int_0^t \Delta p^+(\tau) d\tau > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases} \quad (8)$$

3.3. Extension of RP (2) method based on instantaneous positive sequence

When using p^+ and q^+ in Eq. (2), the 2nd novel extended method is proposed as the following rule (RP (9)):

$$\begin{cases} \text{if } \Delta e_q^+ = \int_0^t \Delta q^+(\tau) d\tau > 0 \text{ and } \int_0^t \Delta \tan(\theta)^+ d\tau = \int_0^t \Delta \left(\frac{q^+}{p^+} \right) d\tau > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases} \quad (9)$$

3.4. Extension of RCC (3) method based on instantaneous positive sequence

By using the instantaneous positive sequence current (i^+) in Eq. (3), the 3rd novel extended method is introduced as the following rule (RCC (10)):

$$\begin{cases} \text{if } \int_0^t \Delta (i^+ \cos(\theta)^+) d\tau > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases} \quad (10)$$

3.5. Extension of SST (4) method based on instantaneous positive sequence

When using the instantaneous positive sequence current and voltage (i^+ and v^+) in Eq. (4), the 4th novel extended method is introduced as the following rule (SST (11)):

$$\begin{cases} \text{if slope } (i^+, |v^+ \cos(\theta)^+|) \text{ line} < 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases} \quad (11)$$

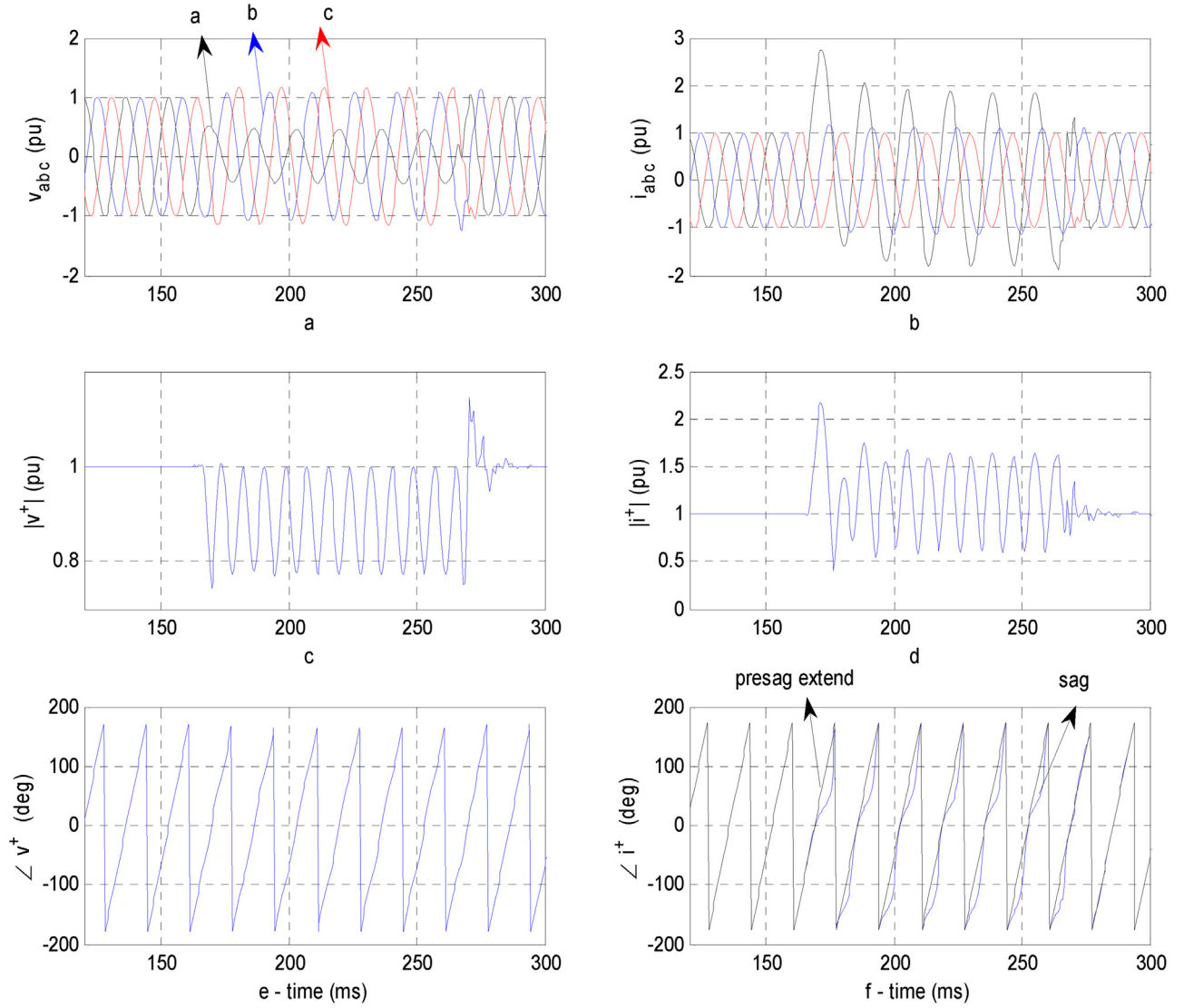


Fig. 2. Voltages and currents for a DS earth fault (a,b) line voltages and currents. (c,d) Magnitude of instantaneous positive sequence voltage and current. (e,f) Phase of instantaneous positive sequence voltage and current.

3.6. Extension of DR (6) method based on instantaneous positive sequence

By using the instantaneous positive sequence impedance ($z^+ = v^+/i^+$) in Eq. (6), the 5th novel extended method is introduced as the following rule (DR (12)):

$$\begin{cases} \text{if } \int_0^t \Delta |z^+| d\tau < 0 \text{ and } \int_0^t \Delta \angle z^+ d\tau > 0 \Rightarrow \text{DS} \\ \text{else} \Rightarrow \text{US} \end{cases} \quad (12)$$

4. Performance evaluation

4.1. Case study

The simulated case study is based on actual data of a Brazilian regional power network, according to the single line diagram shown in Fig. 3. The network consists of long and non-compensated lines, with large distances between load centers and power generation plants. The main generation units are natural gas, hydroelectric plants and sugarcane bagasse-fired thermoelectric plants. This real network has been modeled by PSCAD/EMTDC at a frequency of

60 Hz and the output data are processed via MATLAB codes. Also, the network is highly interconnected and there is a considerable amount of Distributed Generation (DG) units. The features of this network are provided in Table 2.

The monitors are installed in 6 points of the network (lower voltage side of power transformers), with 5 monitors on the boundary between the 230 and 138 kV systems (M1, M2, M3, M4, and M5) and one monitor between the 138 and 13.8 kV system (M6), as shown in Fig. 3. M1 is at the border of a single source radial network with passive RL loads at the DS side. M2 is at the border of a two sources radial network due to the presence of a DG (138 kV, 15 MVA) at the DS side. M3, M4 and M5 are at the border of a two sources interconnected network and M6 is at the border of a single source radial network with active induction motor load (13.8 kV, 3200 HP) at the DS side. In all monitors except M3, the active power flows from the highest voltage level system to the lowest (from left to right in Fig. 3). Therefore, the US side is equivalent to outside and the DS is equivalent to inside when the lowest voltage level system is taken as the reference.

A total of 14 fault points were defined, eight of them (F1, F2, F4, F5, F6, F7, F8 and F14) were located in the 138 kV system, four points (F9, F10, F11 and F12) were simulated in the 230 kV system and,

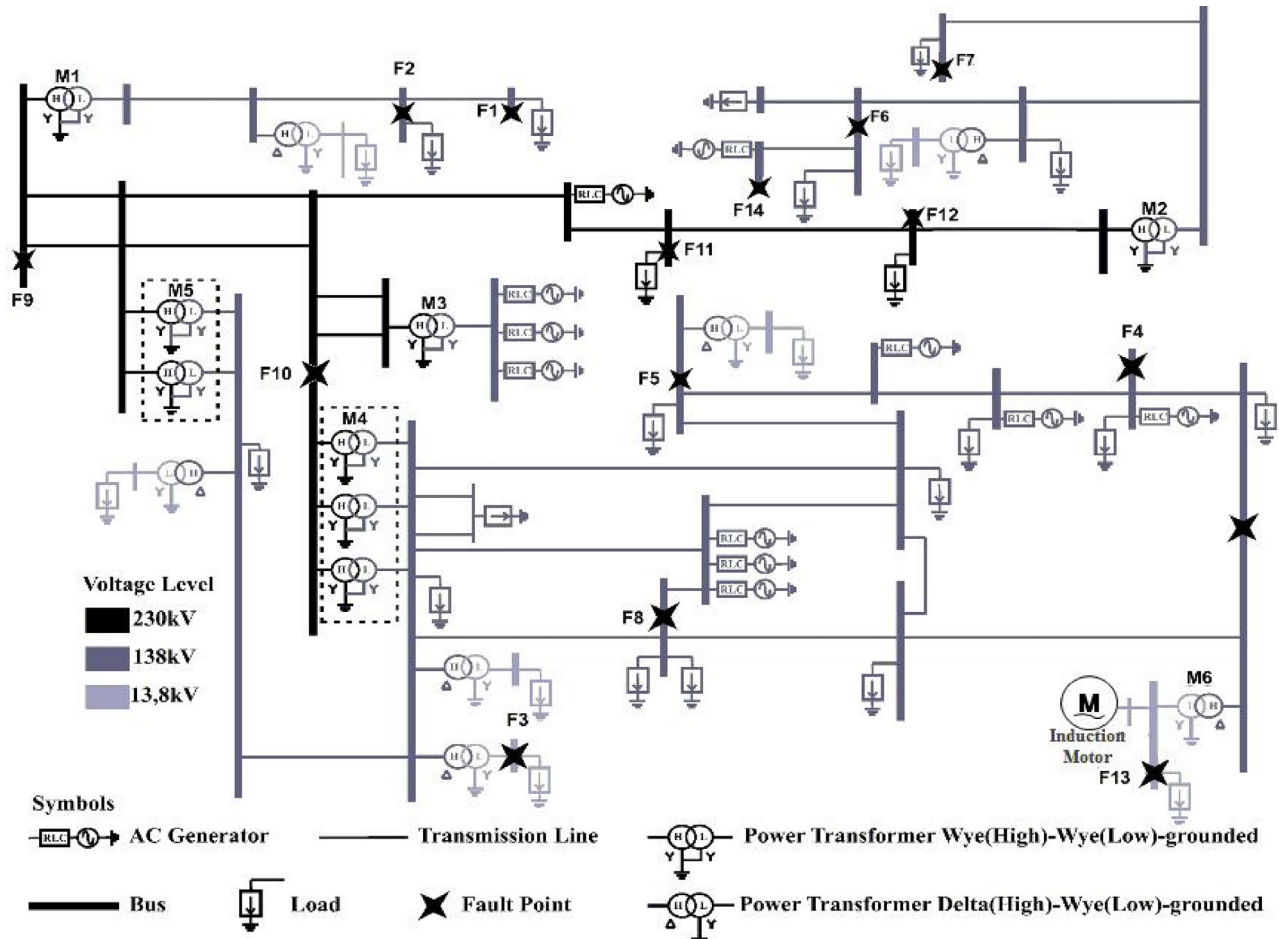


Fig. 3. Single line diagram of a Brazilian regional power network, faults (F1–F14) and monitors (M1–M6) are indicated.

Table 2
Features of a Brazilian regional power network.

Voltage levels	Config.	Generation capacity/present demand	Transformers winding	Transformers capacity (MVA)	No. of transformers	Lines length (km)	No. of lines
Trans. (230 kV), sub trans.(138 kV) and distr. (13.8 kV)	Radial and mesh	1643 MVA/690 MW with high load factor	Y–Y and Y–D grounded	2076	93	6619	67

Table 3
Actual sag source location for all monitors.

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14
M1	↓	↓	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
M2	↑	↑	↑	↑	↑	↓	↓	↑	↑	↑	↑	↑	↑	↓
M3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
M4	↑	↑	↓	↓	↓	↑	↑	↓	↓	↑	↑	↑	↓	↑
M5	↑	↑	↓	↓	↓	↑	↑	↓	↓	↑	↑	↑	↓	↑
M6	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↓	↑

Legend: ↓: DS, ↑: US.

lastly, two points (F3 and F13) were located in the medium voltage system (13.8 kV). Four types of fault were simulated at each point: LLL (symmetrical faults), LG in phase-a (earth faults), LLG and LL in bc (asymmetrical faults) with duration of 100 ms and incident angle of 0°. In all cases, two fault resistance values were used: 0.001 Ω (solid fault or low Rf) and 80 Ω (high Rf). Such combinations yielded 112 fault cases, which were evaluated in the 6 monitors, totaling 672 cases for testing all sag source location discussed and extended

methods, where 256 of them were in the DS direction and 416 in the US direction. Both symmetrical voltage sags (168 examples), and asymmetrical voltage sags (336 examples) were examined, as well as voltage sags due to earth faults (168 examples). Furthermore, voltage sags due to faults with low Rf and high Rf were separately examined (336 examples for each one). Table 3 presents the actual voltage sag source location for each simulated fault.

4.2. Results obtained

The results of simulation of voltage sag source location for methods known from literature as well as for extended methods in certain fault locations and monitors are shown in Figs. 4–6. An US (relative to M6) symmetrical fault with low R_f was generated at F4. The voltages and currents were captured at the terminals of an induction motor (M6 monitor). The obtained results using the methods of DPE (1) based on instantaneous values (denoted by “i”) and DPE (8) based on instantaneous positive sequence values (denoted by “i+”) showed a negative final value for ΔE and Δe_p^+ , respectively, which correctly indicated an US event for both conventional and extended methods of DPE (Fig. 4a and b). Note that the DPE (8) method is an extension of DPE (1).

Let us observe the result obtained by the RCC (3) method based on line phasors (denoted by “Lph.”) and its extended method, i.e. RCC (10) based on instantaneous positive sequence components

during the DS (relative to M1 monitor) LL fault with high R_f caused at F2, as shown in Fig. 4c and d, respectively. Line voltages and currents are captured at M1. In Fig. 4c the results obtained from RCC (3) (denoted by “Lph”) are inconclusive because of opposite signs for $\Delta(I \cos(\theta))$ in phases a, b and c, which shows the method is not suitable for locating asymmetrical faults. These changes in a-phase, b-phase and c-phase are positive, negative and zero, respectively. Fig. 4d shows a positive final value for integral of $\Delta(i^+ \cos(\theta)^+)$ which correctly indicates a DS event for the extended method of RCC (10) by using instantaneous positive sequence values.

A voltage sag due to an LLG fault with high R_f at F7 is considered and captured in the US side of M2. The slope of a-phase is opposite from b and c phases (a-phase slope is positive whereas other phases have negative slopes) in $(I, |V \cos(\theta)|)$ (Fig. 4e, SST (4)), thus, the result is inconclusive. But the slope of $(i^+, |v^+ \cos(\theta)^+|)$ (Fig. 4f, SST

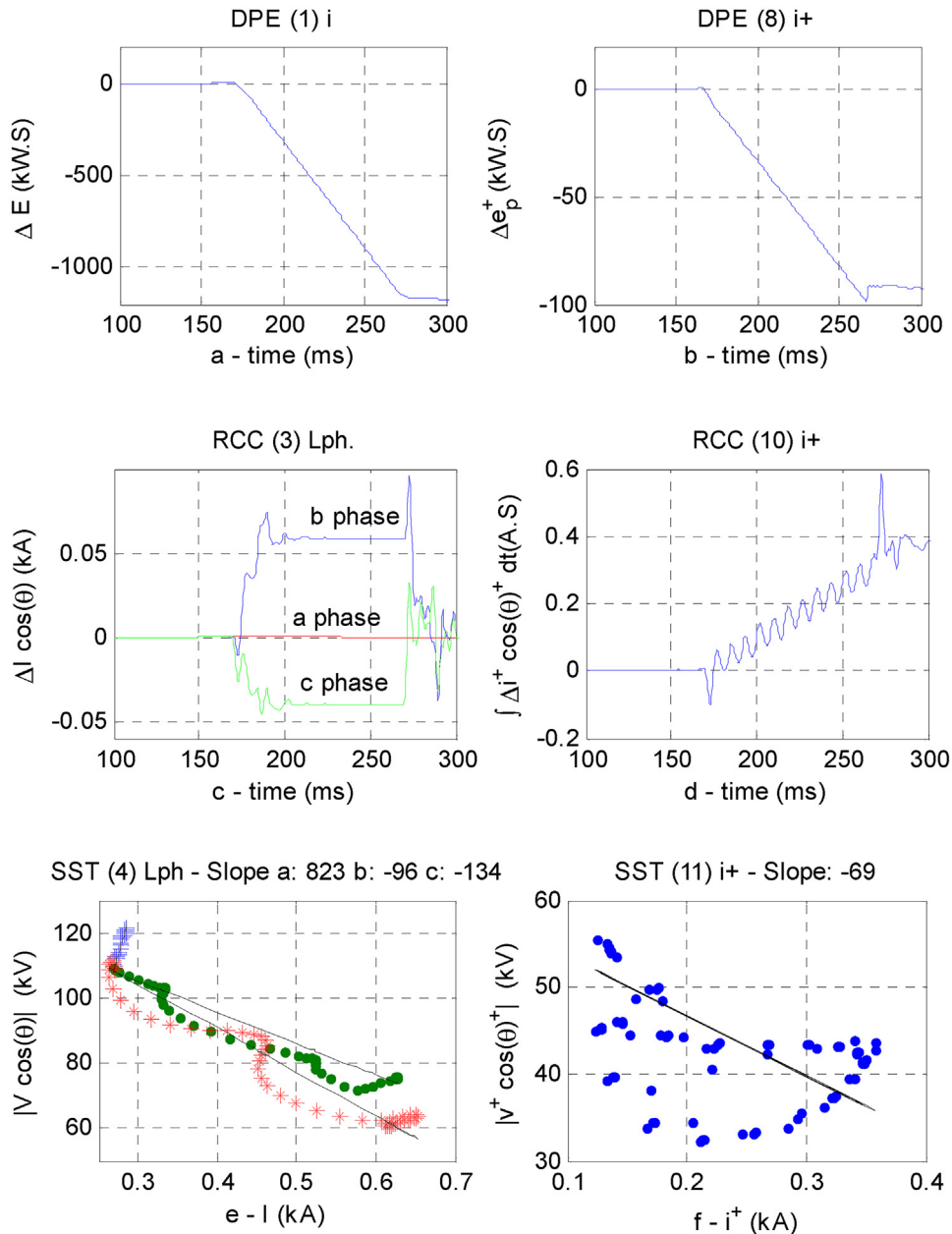


Fig. 4. Results determined for original and extended methods of DPE, RCC and SST (a,b) DPE methods in M6 for US symmetrical fault with low R_f at F4. (c,d) RCC methods in M1 for DS LL fault with high R_f at F2. (e,f) SST methods in M2 for DS LLG fault with high R_f at F7.

(11)) is negative (-69), which shows that the extended method of SST (10) by using instantaneous positive sequence values correctly indicates a DS event.

When the currents and voltages are captured at M3 a DS earth fault with low R_f is caused at the DG source terminals at F14. Fig. 5a shows a positive final value during sag for the changes of reactive power (ΔQ) and a negative value of $\tan(\theta)$ during disturbance (Fig. 5c (red dash lines)), which incorrectly indicates an US fault for the RP (2) conventional method (denoted by “i”). By using the changes as $\Delta \tan(\theta)$ (Fig. 5c (blue solid lines)) which shows positive steady state values, instead of $\tan(\theta)$, the results are improved and the RP (2) improved method (denoted by “i+”) correctly indicates a DS event for F14 DS fault. In the extended method of RP (9), both of the resulting integral values of Δe_q^+ and $\Delta \tan(\theta)^+$ based on instantaneous positive sequence components are positive (Fig. 5b and d). Therefore, RP (9) extended methods correctly indicate a DS event for that fault.

The results obtained in M3 for DR methods based on impedance due to a voltage sag generated by a DS symmetrical fault with high R_f at F5, are shown in Fig. 6. The DR (5) conventional method based on positive sequence values (denoted by “+”) incorrectly indicates a US fault because despite $\Delta |Z^+|$ being negative, $\angle Z$ is negative too (red dash line in Fig. 6b). On the other hand, the method of DR (6) (denoted by “i+”) correctly classifies the location of the fault as DS because $\Delta |Z^+| < 0$ (Fig. 6a) and $\Delta \angle Z^+ > 0$ (blue solid line in Fig. 6c). The final values of integral of $\Delta |z^+|$ (Fig. 6b) and $\Delta \angle z^+$ (Fig. 6d) are negative and positive respectively. This shows that the extended method of DR (12) based on instantaneous positive sequence components correctly locates a DS event.

Altogether, 672 different examples of voltage sags are analyzed, whereas effectiveness is determined for all the methods. Voltage sags due to symmetrical faults, asymmetrical faults and earth faults

at DS/US locations with high/low R_f are considered both separately and collectively and the results obtained are shown in Figs. 7–11 for DPE, RP, RCC, SST and DR methods, respectively. These results show that all extended methods (denoted by “i+”), have better performance than the conventional methods known from literature except the DPE methods, where performances are almost equal. RP (9) and DR (12) methods in US and DS sags, respectively, have a slightly lesser accuracy (about 3% and 5%) relative to RP (5) (for RP (9)) and DR (5) and (6) methods (for DR (12)). Nonetheless, the methods based on line phasors, particularly the SST (3) and RCC (4) methods, show very low effectiveness when the phase quantities (denoted by “Lph.”) are used. The DR (5) and (6) methods based on positive sequence phasors (denoted by “+”) show low effectiveness, which is mostly due to the DS events. Note that the DR (6) that uses change of impedance angle has, in almost all simulated sags, higher accuracy than DR (5), which utilizes impedance angle.

5. Discussion

Sections 5.1–5.4 provide a discussion of the extended new methods and conventional methods.

5.1. Comparison between all methods for all monitors and fault cases

This section presents an evaluation of effectiveness of all methods for all simulated cases, which is performed separately in all monitors. Its results are illustrated in Table 4. Table 4 shows that effectiveness in monitors M3–M5 is generally weaker than in other monitors. All extended methods yield better results than original methods except RP (9) (88%) relative to RP (2) (99%) in M1. The only monitor where DR (6) is less accurate than DR (5) is M1. This is

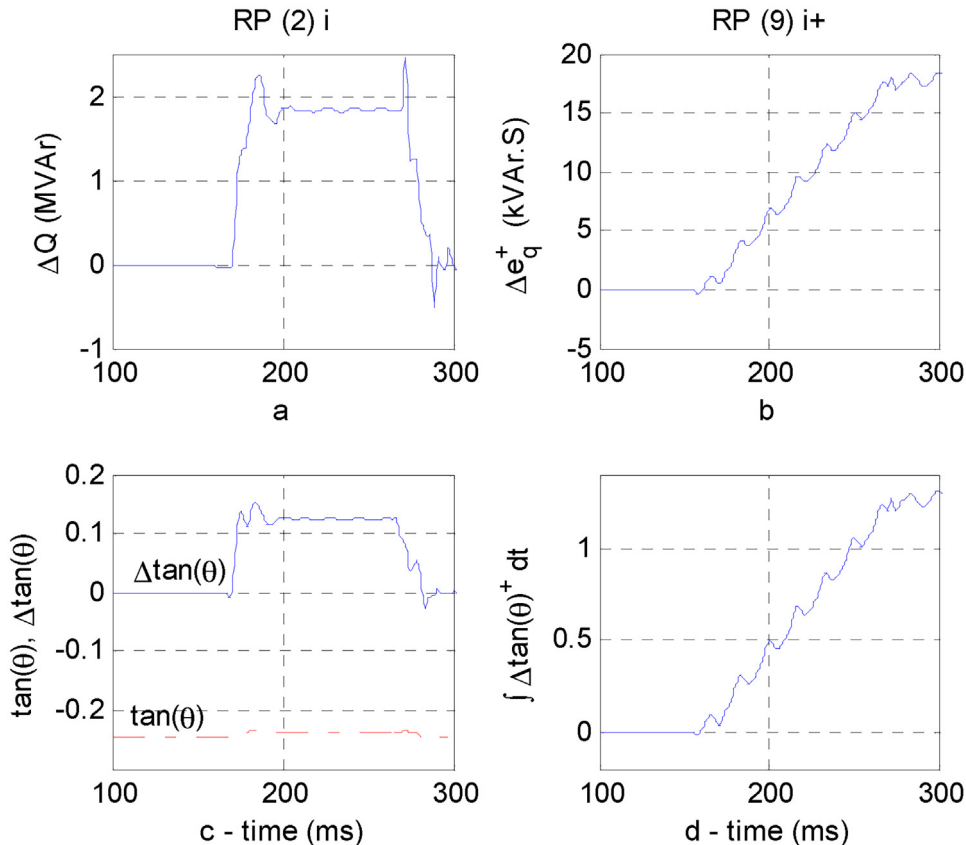


Fig. 5. Results determined in M3 by RP methods for DS earth fault with low R_f at F14 (a,c) RP (2) (b,d) RP (9). (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

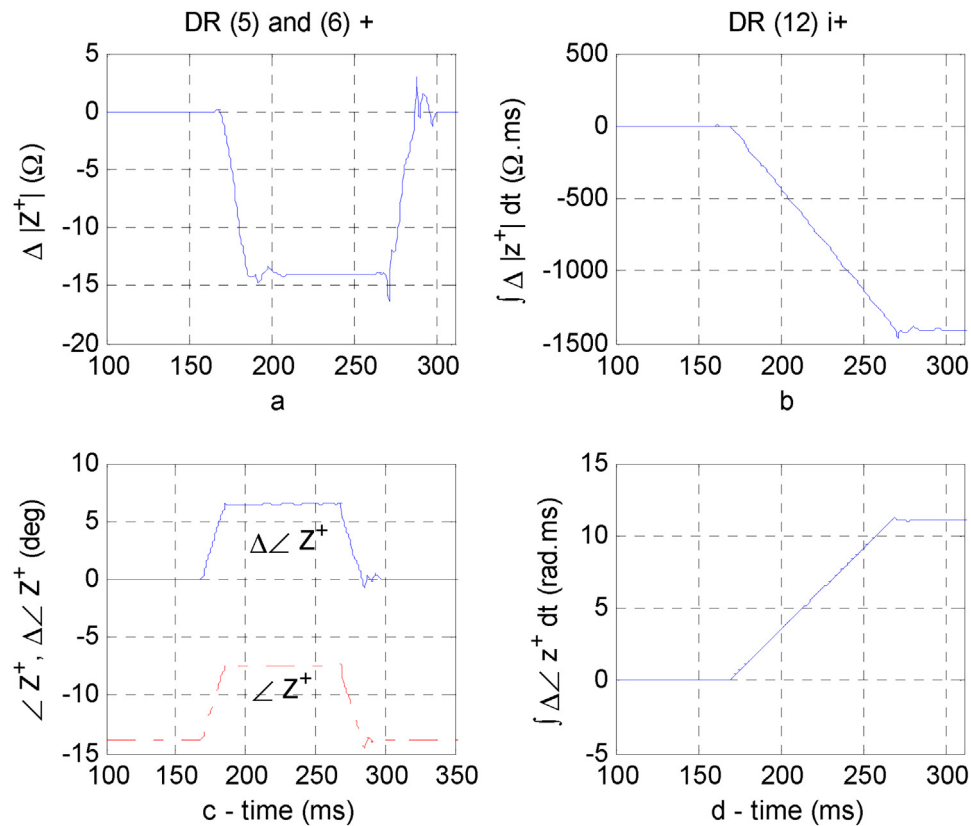


Fig. 6. Results determined in M3 by DR methods for DS symmetrical fault with high Rf at F5 (a,c) with $\angle Z^+$ DR (5); (a,c) with $\Delta \angle Z^+$ DR (6); (b,d) DR (12). (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

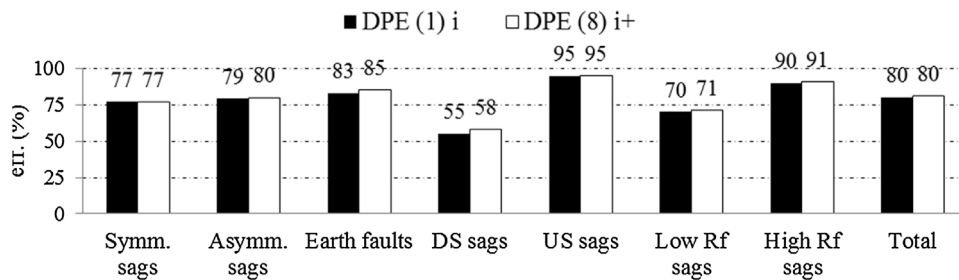


Fig. 7. Performance comparison between the DPE methods for all simulated sags in all monitors.

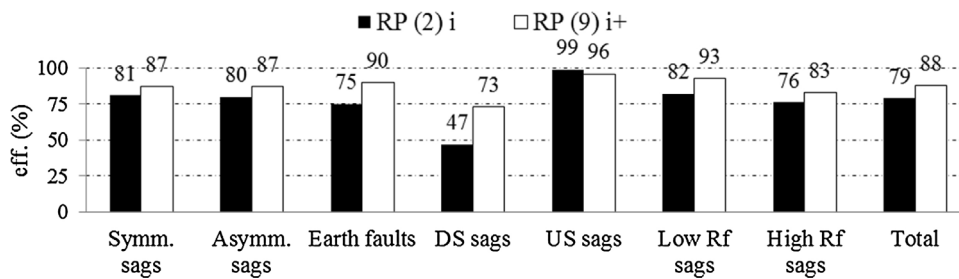


Fig. 8. Performance comparison between the RP methods for all simulated sags in all monitors.

mostly due to the US faults in which the impedance did not change during the sag. Note that M1 is located in the boundary of a one-source radial system. The DR (6) and extended DR (12) methods are the only methods that show correct results in all 672 examples of sags in M2.

5.2. Qualitative evaluation of all methods for all monitors and fault cases

Table 5 qualitatively indicates the comparisons between different methods for all simulated sags in all monitors. The accuracy of methods is divided into four categories: methods with high effectiveness ($\geq 90\%$) (H), methods with acceptable effectiveness

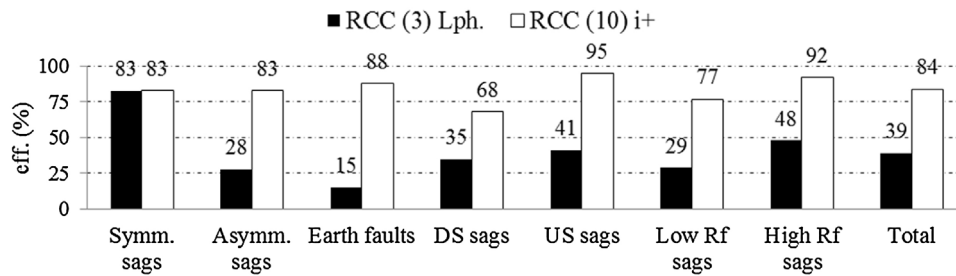


Fig. 9. Performance comparison between the RCC methods for all simulated sags in all monitors.

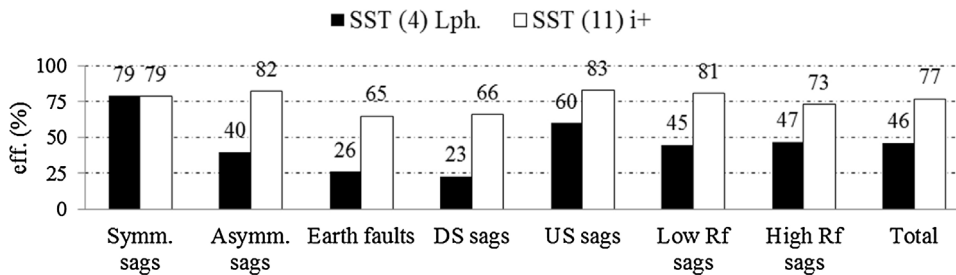


Fig. 10. Performance comparison between the SST methods for all simulated sags in all monitors.

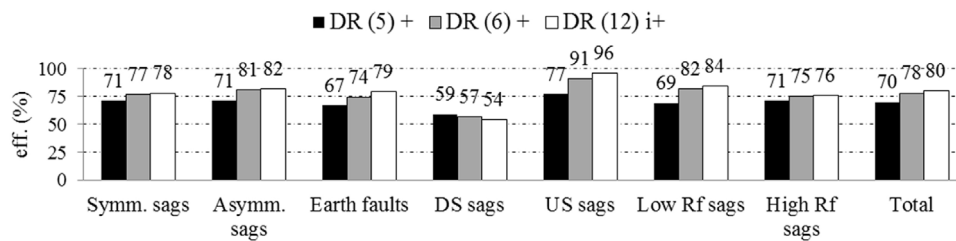


Fig. 11. Performance comparison between the DR methods for all simulated sags in all monitors.

Table 4

Effectiveness of all discussed methods for voltage sag source location for total cases in each monitor.

Method	DPE (%)	RP (%)		RCC (%)		SST (%)		DR (%)		
	(1) i, (8) i+	(2) i	(9) i+	(3) Lph.	(10) i+	(4) Lph.	(11) i+	(5) +	(6) +	(12) i+
M1	95	99	88	43	95	85	80	99	87	89
M2	81	92	98	39	90	62	99	98	100	100
M3–M5	67	61	81	34	73	24	60	67	66	66
M6	97	94	96	51	99	54	96	18	79	90
total	80	79	88	39	84	46	77	70	78	80

Legend: Lph: line phasor values, +: positive sequence phasors, i: line instantaneous values, i+: instantaneous positive sequence components.

Table 5

Qualitative comparisons between different methods for all simulated sags in all monitors.

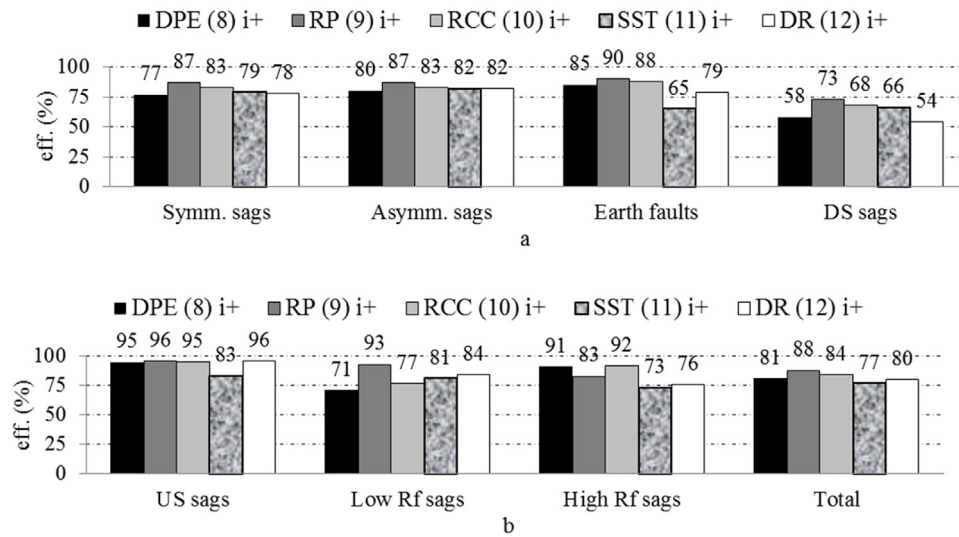
Method	Symm sags	Asymm sags	Earth faults	DS sags	US sags	Low Rf sags	High Rf sags	Total
DPE (1) [7]	A	A	A	Q	H	Q	H	A
DPE (8) [proposed]	A	A	A	Q	H	Q	H	A
RP (2) [14]	A	A	Q	×	H	A	A	A
RP (9) [proposed]	A	A	H	Q	H	H	A	A
RCC (3) [17]	A	×	×	×	×	×	×	×
RCC (10) [proposed]	A	A	A	Q	H	A	H	A
SST (4) [16]	A	×	×	×	Q	×	×	×
SST (11) [proposed]	A	A	Q	Q	A	A	Q	A
DR (5) [22]	Q	Q	Q	Q	A	Q	Q	Q
DR (6) [25]	A	A	Q	Q	H	A	Q	A
DR (12) [proposed]	A	A	A	Q	H	A	A	A

Legend: H—high effectiveness ($\geq 90\%$), A—acceptable effectiveness (76–89%), Q—questionable effectiveness (51–75%), ×—insufficient effectiveness ($\leq 50\%$).

Table 6

Effectiveness of all discussed methods for voltage sag source location against high resistance faults.

Method	DPE (1)	DPE (8)	RP (2)	RP (9)	RCC (3)	RCC (10)	SST (4)	SST (11)	DR (5)	DR (6)	DR (12)
Type	conv.	prop.	conv.	prop.	conv.	prop.	conv.	prop.	conv.	conv.	prop.
Eff. (%)	90	91	76	83	48	92	47	73	71	75	76

**Fig. 12.** Comparison between the five extended methods against: (a) symmetrical sags, asymmetrical sags, earth faults and downstream sags. (b) Upstream sags, sags with low Rf, sags with high Rf and total cases.

(76–89%) (A), methods with questionable effectiveness (51–75%) (Q) and methods with insufficient effectiveness ($\leq 50\%$) ($\times$). As Table 5 shows, the RP (9) extended method has high effectiveness performance (more than 90%) in all cases except in the case of sags with high RF, where the RCC (10) extended method has better performance. All extended methods show better performance than methods known from literature. Also, according to this table, the performance of all methods can be listed from best to worst as:

RP (9), RCC (10), DPE (1), DPE (8), DR (12), DR (6), RP (2), SST (11), DR (5), SST (4) and RCC (3) (extended methods are in bold).

5.3. Comparison between the five extended methods

A comparison between the five proposed methods, i.e. DPE (8), RP (9), RCC (10), SST (11) and DR (12) is shown in Fig. 12. The best performances in each case are:

- Symmetrical sags, RP (9): 87%,
- Asymmetrical sags, RP (9): 87%,
- Earth faults, RP (9): 90%,
- DS sags, RP (9): 73%,
- US sags, RP (9) and DR (12): 96%,
- Sags due to low resistant faults, RP (9): 93%,
- Sags due to high resistant faults, RCC (10): 92%,
- Total cases, RP (9): 88%.

So the method of RP (9) shows the highest accuracy in nearly all cases, except in sags with high fault resistance, where the method with the highest accuracy is RCC (10).

5.4. High resistant faults

Locating high impedance faults (HIF) or high resistance faults (HRF) has been one of the greatest challenges in protection and maintenance of power networks. Also, a fault with current magnitude close to the load current level is not detectable by over-current

relays. Therefore, voltage sags due to such faults are separately investigated in this section. Table 6 shows the effectiveness of all methods for high resistance faults. Because in this type of faults the magnitude of current changes is very low, the conventional method of RCC (3) and SST (4) that use the current changes have the worst performances, whereas the proposed method of SST (4) in this case has the best performance. The methods of DPE that use changes of active power also have very good performance.

6. Conclusion

A presentation and evaluation of eleven different methods for locating voltage sags source has been performed, where five cases were introduced as extended methods using instantaneous positive sequence components. All discussed methods (both known from literature and extended) were extensively tested and comprised applying numerical simulations in a Brazilian power network. Symmetrical, asymmetrical and earth faults with low/high resistance at downstream and upstream locations have been simulated. After studying the results obtained in the network for all monitors, it can be concluded that all the methods are highly effective in cases of symmetrical sags, whereas only the methods that are based on change of power, energy and impedance can be used for locating asymmetrical sags and earth faults. The main contribution of our paper is the finding that in cases of asymmetrical sags and earth faults, the proposed extended methods are much more effective than the conventional phasor-based methods. On the other hand, the methods with the highest effectiveness were the proposed extended methods of RP (9) and RCC (10) in all type faults, while RCC (3) based on line phasors had insufficient effectiveness. The results of our extended methods will be better specified in smaller networks with DGs and active loads, which will be discussed in our next papers.

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References

- [1] M.J.H. Bollen, Understanding Power Quality Problems: Voltage Sags and Interruption, IEEE Press, New York, 2000.
- [2] J.V. Milanovic, C.P. Gupta, Probabilistic assessment of financial losses due to interruptions and voltage sags—part I: the methodology, IEEE Trans. Power Deliv. 21 (2) (2006) 918–924.
- [3] J.V. Milanovic, C.P. Gupta, Probabilistic assessment of financial losses due to interruptions and voltage sags—part II: practical implementation, IEEE Trans. Power Deliv. 21 (2) (2006) 925–932.
- [4] A. Honrubia-Escribano, et al., Influence of voltage dips on industrial equipment: analysis and assessment, Int. J. Electr. Power Energy Syst. 41 (1) (2012) 87–95.
- [5] G. Lei, W. Gu, X. Yuan, A voltage sag index considering compatibility between system and sensitive equipment, Trans. China Electrotech. Soc. 25 (12) (2010) 132–138.
- [6] X. Xiangning, et al., Study on distribution reliability considering voltage sags and acceptable indices, IEEE Trans. Power Deliv. 22 (2) (2007) 1003–1008.
- [7] A.C. Parsons, et al., A direction finder for power quality disturbances based upon disturbance power and energy, IEEE Trans. Power Deliv. 15 (3) (2000) 1081–1086.
- [8] B. Polajžer, G. Štumberger, D. Dolinar, Evaluation of different methods for voltage sag source detection based on positive-sequence components, in: International Conference on Renewable Energies and Power Quality (ICREPOQ09), Valencia (Spain), 15th to 17th April, 2009.
- [9] B. Polajžer, et al., Detection of voltage sag sources based on instantaneous voltage and current vectors and orthogonal Clarke's transformation, IET Gener. Transm. Distrib. 2 (2) (2008) 219–226.
- [10] W. Kong, X. Dong, Z. Chen, Voltage sag source location based on instantaneous energy detection, Electr. Power Syst. Res. 78 (11) (2008) 1889–1898.
- [11] A. Mohamed, H. Shareef, M.F. Faisal, A novel voltage sag source detection method using TFR technique, Int. Rev. Electr. Eng. 5 (5) (2010) 2301–2309.
- [12] X. Zhang, Y.H. Xu, Analysis of voltage sag source location based on wavelet-multiresolution method, 2010 Asia-Pacific Power and Energy Engineering Conference (2010).
- [13] W. Ling Ai, H. Shareef, A single monitor method for voltage sag source location using Hilbert–Huang transform, Res. J. Appl. Sci. Eng. Technol. 5 (1) (2013) 192–202.
- [14] R.C. Leborgne, R. Makaliki, Voltage sag source location at grid interconnections: a case study in the Zambian system, Power Tech, 2007 IEEE Lausanne (2007).
- [15] F.O. Passos, et al., An alternative approach to locating voltage sag source side at the point of common coupling based on power-flow information, J. Control Autom. Electr. Syst. 26 (5) (2015) 579–587.
- [16] C. Li, et al., Method for voltage-sag-source detection by investigating slope of the system trajectory, IEE Proc.: Gener. Transm. Distrib. 150 (3) (2003) 367–372.
- [17] N. Hamzah, A. Mohamed, A. Hussain, A new approach to locate the voltage sag source using real current component, Electr. Power Syst. Res. 72 (2) (2004) 113–123.
- [18] T. Tayjasananant, L. Chun, W. Xu, A resistance sign-based method for voltage sag source detection, IEEE Trans. Power Deliv. 20 (4) (2005) 2544–2551.
- [19] W. Kanokbannakorn, T. Saengsuwan, S. Sirisukprasert, Unbalanced voltage sag source location identification based on superimposed quantities and negative sequence, 8th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON) (2011).
- [20] K. Zhu, et al., Voltage sag source location technology based on corresponding sequence components, IET Gener. Transm. Distrib. 9 (9) (2015) 820–827.
- [21] T. Yi, H. Jie, L. Hao, Method for voltage sag source location based on the internal resistance sign in a single-port network, IET Gener. Transm. Distrib. 10 (7) (2016) 1720–1727.
- [22] A.K. Pradhan, A. Routray, Applying distance relay for voltage sag source detection, IEEE Trans. Power Deliv. 20 (1) (2005) 529–531.
- [23] M.H. Moradi, Y. Mohammadi, A new current-based method for voltage sag source location using directional overcurrent relay information, Int. Trans. Electr. Energy Syst. 23 (2) (2013) 270–281.
- [24] M.H. Moradi, Y. Mohammadi, M. Hosseini Tayyebi, A novel method to locate the voltage sag source: a case study in the Brazilian power network (Mato Grosso), Przegląd Elektrotechniczny (Electr. Rev.) 88 (3b) (2013) 112–115.
- [25] Y. Yilin, X. Yonghai, Research of method for voltage sag source detection in power distribution network, 2011 6th IEEE Conference on Industrial Electronics and Applications (2011).
- [26] J. Gomez, et al., A novel methodology to locate originating points of voltage sags in electric power systems, 8th International Conference and Exhibition on Electricity Distribution, 2005. CIRED (2005).
- [27] R.C. Leborgne, D. Karlsson, Voltage sag source location based on voltage measurements only, Electr. Power Qual. Util. 14 (1) (2008) 25–30.
- [28] A.K. Pradhan, A. Routray, S.M. Gudipalli, Fault direction estimation in radial distribution system using phase change in sequence current, IEEE Trans. Power Deliv. 22 (4) (2007) 2065–2071.
- [29] M.H. Moradi, Y. Mohammadi, Voltage sag source location: a review with introduction of a new method, Int. J. Electr. Power Energy Syst. 43 (1) (2012) 29–39.
- [30] R.C. Leborgne, D. Karlsson, J. Daalder, Voltage sag source location methods performance under symmetrical and asymmetrical fault conditions, in: 2006 IEEE/PES Transmission & Distribution Conference and Exposition, Latin America, 2006.
- [31] B. Polajžer, et al., Generalization of methods for voltage-sag source detection using vector-space approach, IEEE Trans. Ind. Appl. 45 (6) (2009) 2152–2161.
- [32] B. Polajžer, G. Štumberger, D. Dolinar, Detection of voltage sag sources based on the angle and norm changes in the instantaneous current vector written in Clarke's components, Int. J. Electr. Power Energy Syst. 64 (2015) 967–976.
- [33] B. Polajžer, et al., Instantaneous positive-sequence current applied for detecting voltage sag sources, IET Gener. Transm. Distrib. 9 (4) (2015) 319–327.
- [34] S. Zhenguo, P. JinPing, K. Jian, Locating voltage sag source with impedance measurement, Power System Technology, International Conference (2010).
- [35] V.B. Núñez, et al., Evaluation of fault relative location algorithms using voltage sag data collected at 25-kV substations, Eur. Trans. Electr. Power 20 (1) (2010) 34–51.
- [36] Y. Mohammadi, M.H. Moradi, R.C. Leborgne, A novel method for voltage-sag source location using a robust machine learning approach, Electr. Power Syst. Res. 145 (2017) 122–113623.
- [37] A.D. Santosa, M.T.C.D. Barrosa, Voltage sag prediction for network planning, Electr. Power Syst. Res. 140 (2016) 976–983.
- [38] M. Sengul, The effects of the voltage recovery after the voltage sag on gas discharge lamps, Electr. Power Syst. Res. 136 (2016) 309–321.
- [39] S. Khoshkbar, K.M. Smedley, Fast and precise voltage sag detection method for dynamic voltage restorer (DVR) application, Electr. Power Syst. Res. 130 (2016) 192–207.
- [40] G.C. Paap, Symmetrical components in the time domain and their application to power network calculations, IEEE Trans. Power Syst. 15 (2) (2000) 192–207.